SAINT MICHAEL COLLEGE OF CARAGA

IMPACT OF OPLAN LIGTAS NA PAMAYANAN ON FIRE SAFETY BEHAVIOR OF HOUSEHOLDS IN BUENASCAR, AGUSAN DEL NORTE

A Thesis Presented To
The Faculty of
College of Criminal Justice Education

In Partial Requirements for the Degree
BACHELOR OF SCIENCE IN CRIMINOLOGY

By

RACHELLE ANN D. PALER
MARK LOUIS A. PILAPIL
KIM BRYLL E. CUPAY
EMMANUEL G. JUMAWID
BERNABE C. BUAL, JR.
URIEL CLARENCE P. ELIOT

ADVISER: MS. JILL M. MESA, MSCRIM (CAR)

OCTOBER 2025

CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the Study

Fire incidents continue to be a serious threat to people's safety in many Filipino homes. Among the causes are poor electrical wiring, a lack of understanding about fire safety, and the use of lightweight or easily flammable materials in home construction. Unprepared households are especially at risk because even small fires can cause severe property damage, fatalities, and lasting social and economic problems for families and local governments (Paulino & Matthew Paulino, 2023).

The Bureau of Fire Protection (BFP) created and carried out Oplan Ligtas na Pamayanan (OLP), a nationwide community-based fire prevention program, to raise awareness of fire safety issues and promote safer barangay practices in response to these threats. Through community fire safety education, house-to-house inspections, and residents' active involvement in various risk-reduction initiatives, the program seeks to lower the risk of fire (BFP, 2022). The program was basically focused on urging households to use proactive fire safety precautions.

The program aimed to encourage households to adopt proactive fire safety measures, including proper handling and safe storage of flammable materials, correct use of electrical appliances, and preparedness for emergencies (Capiña, 2022); Macarubbo, 2022). OLP has become a vital strategy for lowering the incidence and consequences of fires in various Philippine communities over time (Lagumbay et al., 2023). Currently, no empirical evidence exists on how OLP affects household fire safety practices in smaller communities, particularly in Buenavista, Nasipit, and Carmen, Agusan del Norte. This lack of local data matters because community programs may function differently depending on participation, income, and local risk factors.

By investigating the effects of Oplan Ligtas na Pamayanan on fire safety practices in homes in Buenavista, Nasipit, and Carmen, Agusan del Norte, this study seeks to close the gap. The study will give the Bureau of Fire Protection, local government entities, and community leaders useful information by evaluating the program's Efficacy in homes. The researchers expect that the results will help guide policy decisions and improve fire management systems, ensuring that fire safety practices in the community are strong and lasting.

The Fire and Disaster Management Agency (FDMA) in Japan is responsible for fire safety and emphasizes neighborhood-based initiatives. These include information campaigns, public drills, volunteer organizations for disaster prevention, and routine inspections (FDMA, 2024) International organizations have praised Japan's disaster management systems for their practicality and organization (GFDRR, 2013). Similarly, South Korea's National Fire Agency (NFA) manages various fire safety initiatives, including public awareness campaigns, inspections, and training. Meanwhile, China's fire safety system has developed rapidly with the country's industrialization and urbanization, and recent studies indicate that fire regulations and management practices have improved (Zhang et al., 2019) . The National Fire Fighting Corps (NFFC) implements fire safety measures, including routine inspections and public education campaigns (Zhang, 2021); (Liu et al., 2023). These examples show that community-based approaches are an important part of successful fire prevention worldwide.

The Philippines' tropical climate, especially the high temperatures during the dry season, increases the risk of fire. Dense residential areas and rapid urbanization further increase the country's vulnerability to fire-related disasters (Manalo et al., 2022). Fires often destroy homes and property, cause fatalities, create economic problems, and slow down community development. Because of this, there is a pressing need to improve fire safety and preparedness efforts across the nation. (Ann et al., 2022)

Many households still do not follow basic fire safety measures, even though fire safety programs exist. While significant buildings usually have sprinklers and other safety systems, some homes do not have basic fire equipment, like fire extinguishers (Albano, 2024). Securing a home's electrical wiring reduces the risk of fire. However, many homeowners attempt to save money by doing their own electrical work, which often leads to unsafe or poorly connected wiring.

Surveys conducted in Bislig City supported the findings of (Martin's Electrical and Lighting Company, 2025), which revealed that many households used incorrect fuse ratings, neglected to have their wiring inspected regularly, did not adhere to clear fire safety procedures, and had little knowledge of how to prevent electrical fires. Concrete homes were more likely to have fire extinguishers, proper permits, and better safety practices (Nocete & Enteria, 2020).

According to specific research, being aware of fire safety precautions does not guarantee that people follow them. According to one review, although many homes have fire safety equipment, people often neglect it or are unsure how to use it in an emergency Senthilkumaran et al., (2019). Household negligence and a lack of accountability can reduce the effectiveness of safety measures. Age-related differences in perceptions of safety are also significant. For example, younger people are more careful about fire safety, while older people often take risks, such as leaving appliances on overnight (Aliesa Mohd Zul et al., 2024).

Several studies have examined the implementation and effectiveness of Oplan Ligtas na Pamayanan in different areas of the Philippines. Findings from these studies show that OLP is effective in increasing community awareness of fire hazards and improving residents' knowledge of fire prevention practices. A study published in the *International Journal of Environmental Sciences* reported that regular fire safety education, house-to-house surveys, and active community participation under OLP help reduce fire risks (Abella et al., 2021). Similarly, Garcia (2020) found that municipalities with a higher level of OLP implementation showed better fire

preparedness, although challenges such as limited manpower and lack of equipment affected full implementation.

At the local level in Agusan del Norte, particularly in Buenavista, Nasipit, and Carmen, Oplan Ligtas na Pamayanan is actively implemented through various fire prevention activities conducted by the Bureau of Fire Protection. In Nasipit, Agusan del Norte, the BFP has conducted house-to-house fire safety surveys, fire truck visibility operations, and fire safety information drives as part of OLP implementation. These activities focus on identifying fire hazards inside households and educating residents on proper fire prevention practices. Such efforts support the OLP objective of strengthening direct community engagement and proactive fire risk reduction (Bureau of Fire Protection–Nasipit Fire Station, 2024).

In Carmen, Agusan del Norte, the Bureau of Fire Protection continues to implement Oplan Ligtas na Pamayanan through fire safety awareness campaigns and community education activities. These activities include reminding residents about safe electrical use, fire prevention tips, and emergency response procedures. The implementation of these activities shows the active participation of the BFP Carmen Fire Station in promoting fire safety awareness at the barangay level, which aligns with the goals of OLP to empower communities and reduce fire-related incidents (Bureau of Fire Protection–Carmen Fire Station, 2024).

In Buenavista, Agusan del Norte, Oplan Ligtas na Pamayanan is integrated into the general fire prevention and safety programs of the local BFP. The fire station conducts fire safety education, coordinates with barangay officials, and participates in community fire safety activities to ensure residents are informed and prepared. These initiatives follow the national guidelines of the Bureau of Fire Protection and reflect the municipality's commitment to implementing OLP as part of its local disaster preparedness and public safety strategy (Bureau of Fire Protection–Buenavista Fire Station, 2024).

This study aims to understand fire safety behavior better and to highlight how the Bureau of Fire Protection's awareness programs help increase fire safety knowledge and preparedness in the community. The study examines explicitly how Oplan Ligtas na Pamayanan (OLP) affects home fire safety procedures, emphasizing participation in risk reduction, awareness, and preparedness. The study will ascertain how community-based initiatives, such as inspections, lectures, drills, and risk assessments, affect household behavior in reducing fire hazards by assessing the program's Efficacy.

Several groups could benefit from the study's conclusions. The findings can help households increase their awareness and practical understanding of fire safety. Barangay officials can use the information to improve their community safety initiatives and information campaigns. The Bureau of Fire Protection will use evidence-based insights to improve current programs and policies. Additionally, this study will add to the body of knowledge regarding community-based fire prevention initiatives and their importance for disaster preparedness in local communities.

1.2 Theoretical Framework

This study is anchored on the Protection Motivation Theory (PMT) developed by Rogers (1975) and later refined by Rogers (1983). PMT explains how individuals are motivated to adopt protective behaviors when they are exposed to information about threats and recommended coping strategies. The theory has been extensively applied in studies related to health promotion, disaster preparedness, and safety behavior, making it appropriate for examining household fire-safety behavior (Rogers, 1975; Rogers, 1983; Maddux & Rogers, 1983). The Protection Motivation Theory (PMT) states that when households believe the program's safety measures are effective, feel confident in implementing them even in the face of challenges, and view fire as a serious and likely threat, they are more likely to adhere to fire safety practices. PMT is a helpful way to understand and predict fire safety behavior in households targeted by Oplan Ligtas na Pamayanan.

Studies in the Philippines support this framework, showing that self-efficacy and response efficacy are strong predictors of disaster preparedness (Kurata et al., 2023).

The relevance of PMT to this study lies in its ability to explain how fire safety programs, such as Oplan Ligtas na Pamayanan, influence households' decisions to adopt preventive behaviors. Since the program focuses on increasing awareness, improving skills, and strengthening confidence in fire prevention measures, PMT provides a theoretical foundation for understanding how these interventions motivate households to practice fire safety. PMT posits that protective behavior results from a cognitive evaluation process involving threat appraisal and coping appraisal (Rogers, 1975; Floyd et al., 2000). Threat appraisal involves assessing the severity of a threat and one's vulnerability to it, while coping appraisal involves evaluating the effectiveness of the recommended response and one's perceived ability to perform the behavior successfully (Maddux & Rogers, 1983; Rogers, 1983).

According to PMT, individuals are more likely to engage in protective behavior when they perceive a threat as serious and likely, believe that the recommended actions are effective, and feel confident in their ability to carry out these actions Floyd et al., (2000); Grothmann & Reusswig, (2006).

The key constructs of PMT applied in this study include perceived severity, perceived vulnerability (risk perception), response efficacy, self-efficacy, and attitude toward protective behavior. Perceived severity refers to households' assessment of the potential consequences of fire incidents, while perceived vulnerability reflects their belief regarding the likelihood of experiencing a fire (Rogers, 1975; Rogers, 1983). Response efficacy refers to the belief that the fire safety measures promoted by Oplan Ligtas na Pamayanan are effective in preventing or reducing fire damage, whereas self-efficacy refers to households' confidence in their ability to implement these measures (Bandura, 1977; Maddux & Rogers, 1983).

Protection Motivation Theory is relevant to this study because it explains how the implementation of Oplan Ligtas na Pamayanan influences household fire-safety behavior through changes in perception, confidence, and motivation. Program activities such as house-to-house fire safety surveys, zonal interventions, lectures, and drills increase households' awareness of fire risks and strengthen their belief in the effectiveness of fire prevention strategies. These interventions directly correspond to the threat appraisal and coping appraisal processes described in PMT (Rogers, 1983; Grothmann & Reusswig, 2006).

Empirical studies have demonstrated that PMT is effective in explaining disaster preparedness and safety behaviors, with self-efficacy and response efficacy consistently identified as strong predictors of protective action (Floyd et al., 2000; Grothmann & Reusswig, 2006; Kurata et al., 2023).

According to the theory, being willing to alter one's behavior is the best defense. Researchers define threat appraisal as people's awareness of risk factors, specifically their perceived vulnerability to and the severity of a disaster. These evaluations combine to form Protection Motivation, the mental motivation to practice fire safety.

According to Rogers, (1975) and Floyd et al., (2000) These assessments in PMT are multiplicative: even if residents have strong beliefs, their motivation will decrease dramatically if any one of them is weak.

1.3 Review of Related Literature

In the Philippines, fire incidents continue to pose a significant threat, causing fatalities, property damage, and livelihood disruptions. The BFP has responded by implementing national fire awareness campaigns, including school-based fire drills designed to minimize injuries during emergencies (Opiana, 2017).

Given its direct impact on property and human lives, ensuring fire safety is a top priority (Gautami et al., 2020). In response to the rising number of fire incidents across the country, the Bureau of Fire Protection (BFP) launched a comprehensive awareness campaign in 2019 to commemorate Fire Prevention Month, as required by Presidential Proclamation No. 115-A. The program aimed to establish fire-safe communities that would enhance each Filipino's overall quality of life. (BFP, 2022)

Establishing efficient fire safety procedures is crucial to protecting people and property in a nation highly susceptible to hazards. Beyond being legally mandated, following the Philippine Fire Code is also morally important to save lives and property (Lagata et al., 2022). Effective fire prevention requires regular risk assessments, reliable detection and suppression systems, and proper maintenance of fire safety equipment. Community education and public awareness are equally essential in preventing fire incidents (Fu et al., 2024). Fire risk assessments involve identifying vulnerabilities, recognizing potential hazards, and implementing precautions to reduce risks. By being proactive, both individuals and organizations can prevent fires before they start.

The BFP divides OLP into four categories: "Lingap" (Protective Care), "Handang Pamayanan" (Ready Community), "Bayanihan" (Cooperative Undertaking), and "Kaagapay" (Companion) (BFP, 2022). The Handang Pamayanan program helps communities develop hazard maps and fire safety plans. In contrast, the Bayanihan program trains volunteer brigades to join the Community Fire Auxiliary Group (CFAG) and supports neighborhood projects. The Kaagapay program fosters skill development through barangay visits and home fire safety exercises. The Lingap program honors barangays that conduct thorough house-to-house fire safety surveys, especially in densely populated, high-risk areas (BFP, 2022). Together, these programs encourage planning, readiness, and community involvement, effectively reducing the number of fire incidents (Rosales, 2019).

While BFP teams are en route, the CFAG, a formal volunteer organization composed of locals trained through the Bumbero sa Barangay program, responds to local fire incidents. This group plays a crucial role in enhancing neighborhood safety and strengthening barangay firefighting capabilities. Managed by local officials, the program ensures sustainability and efficient coordination across all aspects of fire response. According to Muico et al. (2024), communities demonstrate a strong commitment to fire safety, with residents actively learning and adhering to safety precautions, highlighting the Efficacy of fire safety education initiatives.

Global best practices emphasize awareness and preparedness as key to reducing fire risks. The University of Texas at Austin promotes fire safety by teaching safe cooking techniques, proper fuel handling, correct use of alarms, and escape route planning. Since residents may have as little as two minutes to evacuate during a fire, the United Ways of New Jersey urges families to practice their fire escape plans regularly. According to Addai et al., (2016) Unsafe electrical wiring by unskilled artisans in Ghana often causes short circuits and fire incidents. Open flames from lanterns, stoves, and cigarettes also remain primary sources of ignition. These incidents underscore the need for improved infrastructure, stricter safety regulations, and more effective global awareness campaigns.

Attitude

Their attitudes toward fire safety consistently influence people's intentions and actual preparedness behaviors. Research has demonstrated that people are more likely to take home safety actions, such as installing alarms or creating escape routes, when they have positive attitudes influenced by social influences, past knowledge, and the perceived value of safety precautions. Along with risk perception and self-efficacy, research on fire safety behavior also finds that attitude is a critical component. (V. Cvetkovic, 2021).

According to the Bureau of Fire Protection (BFP), unsafe LPG use, unattended cooking, and defective wiring are common causes of fire. The correct information and safe procedures can reduce these risks. Despite laws and programs, many people, particularly business owners, are poorly informed about fire safety measures, thereby increasing the risk of fire-related losses (Nudzor, 2023).

Furthermore, residents often take action to put out fires before authorities arrive, showing the importance of community involvement in fire management (Ardani & Permana, 2019). Working together helps connect and empower everyone involved in fire prevention, allowing fires to be managed more effectively and efficiently (Menya & K'akumu, 2016). According to research, attitudes, confidence, and risk awareness are among the ways fire safety knowledge influences household behavior and functioning. Specifically, (Ferrer, 2021) discovered that some towns had high ratings for their fire prevention programs; however, research rarely examines whether officials' expertise or other local factors are responsible for these outcomes. This disparity is significant because new research shows that community involvement and local leaders' technical and procedural expertise are both necessary for successful disaster preparedness, the United Nations Office for Disaster Risk Reduction.

Additionally, (Gandawali & Abdul, 2017) showed a strong correlation between respondents' knowledge of fire safety and the execution of the Oplan Ligtas na Pamayanan program. The Bureau of Fire Protection created an enhancement program to support and reinforce current fire prevention efforts. The study also demonstrated that greater awareness was associated with higher levels of program implementation, underscoring the program's beneficial impact on knowledge of fire prevention.

According to Kurata et al., (2023) Prior knowledge, perceived severity, vulnerability, and personal experiences all influence people's attitudes. People with more positive attitudes are more likely to value prevention programs and take part in safety measures.

Similarly, Muico et al., (2024) evaluated home fire prevention awareness and practices in Barangay San Roque, Municipality of Maco. The findings demonstrated a high level of comprehension and adherence to fire safety precautions, including installing safety equipment and preparing the home for fire emergencies. The study indicates strong engagement in fire safety procedures and a favorable attitude toward fire prevention, despite not being conducted over an extended period. This favorable attitude likely resulted from the BFP's education and awareness initiatives.

In the Philippines, Kurata et al., (2023) found that people's views on the effectiveness of safety programs are closely linked to their positive attitudes toward fire prevention, especially when they understand the risks of fire. Their results provide evidence that awareness-raising, community drills, and active home participation programs, such as OLP, can lead to behavioral and attitudinal changes. This aligns with the Protection Motivation Theory, which holds that when people are aware of the seriousness of a threat and the effectiveness of preventive measures, they are more inclined to take protective actions. (Rogers, 1975).

Self-efficacy

People with special needs and vulnerabilities, such as children, the elderly, people with disabilities or medical conditions, and those who are underserved or have limited resources, have been demonstrated to be better prepared for disasters when they have self-efficacy (Hamann et al., 2016); (Marceron & Rohrbeck, 2019); (Rivera, 2020). Additionally, (Adams et al., 2017) found that self-efficacy can influence the relationship between preparedness and perceptions of health. According to (Marceron & Rohrbeck, 2019) Self-efficacy strengthens people's perceptions of threats, thereby increasing the likelihood that they will adopt disaster preparedness measures.

Research also showed that there is no meaningful connection between threat perception and preparedness for disasters when self-efficacy is low. In other words, if people lack self-efficacy, they are less prepared even when they see a high level of threat. Self-efficacy is often lower in families with members who are especially vulnerable.

(Guevara, 2025) defines fire safety as the steps, rules, and methods used to prevent fires, lower the chances of fire-related incidents, and protect people and property if a fire occurs. To stop fires and lessen their effects, a mix of awareness, readiness, and doable safety precautions is needed. Providing fire extinguishing equipment, such as sprinklers, smoke detectors, fire alarms, and fire extinguishers, is part of being prepared to respond to fire disasters. These efforts decrease potential risks, protecting people more effectively. It also means making sure residents know the fire department's phone numbers and have the skills to use the equipment and alarms. Checking that fire exits are easy to reach during a fire is a common way to measure a building's safety, (Adeleye et al., 2020).

Only when people have reasonable expectations of their capacity to carry out the act will they develop intentions to prepare for disasters (Adams et al., 2019). The majority of research has shown that high self-efficacy can drive specific protective measures in emergencies, like emergency evacuation (Tang & Feng, 2018); (Yu et al., 2020), as well as disaster preparedness intentions or actual behaviors (Samaddar et al., 2014).

Risk Perception

Age and education, as well as fire risk perception and knowledge, were found to be significantly correlated in a study examining the effects of several sociodemographic factors, including age, education level, gender, and prior fire experience (Chan et al., 2018). However, (Kinateder et al., 2015) state that risk perception is an individual's subjective assessment of the likelihood of a fire and the potential severity of its effects, including feelings of vulnerability and

danger, as well as emotional responses. People are more likely to take preventive steps, such as checking electrical systems or keeping fire extinguishers ready, if they think a fire is severe and likely to happen.

In a similar vein, (Kurata et al., 2023) found that perceived severity and vulnerability directly influence Filipinos' fire safety practices, two key components of risk perception. They demonstrate that people are more likely to take preventative measures when they are aware of the risks posed by fire.

Expanding on this notion, theories such as the Protection Motivation Theory (PMT) describe how people are motivated to take protective measures based on their perceptions of risk. Perceived susceptibility—the likelihood of being affected—and perceived severity—the gravity of the consequences—are the two primary components of risk perception, according to these models. These components affect a person's intention to implement safety measures when paired with other elements, such as self-efficacy and belief in the Efficacy of the response (R. A. Ferrer & Klein, 2015). People are more likely to take precautions when they perceive a high risk, but they are also more likely to act carelessly or in an unsafe manner when they perceive a low risk. Research on fire and disaster safety found that residents' views of their risk of fire injury are closely linked to their past fire experiences and their trust in the city's fire response services (Nilson & Mankell, 2022). This suggests that residents should install fire extinguishers, inspect electrical systems, and take other preventive measures to reduce fire risks, and that households that understand the gravity and likelihood of fire hazards are more likely to participate in community fire drills.

Knowledge by itself does not always translate into safe behavior, according to empirical data. When people feel that they are in danger, they are more likely to take action. For instance, (Mccaffrey, 2015) found that when families felt more personally at risk, they were more likely to

keep smoke alarms operational and make evacuation plans. Barangays that actively implemented OLP demonstrated higher levels of awareness and readiness. Still, disparities remained, impacted by residents' perceptions of their susceptibility to fire risks, (Nanta & Pisueña, 2025). Even with information, some households did not buy fire extinguishers or unplug appliances before leaving, because they believed a fire was unlikely. These results imply that information alone is insufficient. To promote the adoption of workable safety measures, effective risk communication must also account for people's attitudes.

House-to-House Fire Safety Survey

The primary method for carrying out zonal interventions is the house-to-house fire safety survey, which the OLP Lingap Program requires. Under the direction of designated Bumbero sa Barangay staff, survey teams assess each home's compliance with fire safety regulations, identify structural and electrical hazards, and offer suggestions for prompt remedial measures. Coordination with barangay officials, recording of hazards found, and submission of combined reports to the BFP District Office for oversight and assessment are all part of each survey.

The Lingap Program's house-to-house fire safety surveys are another way that the OLP emphasizes the value of routine inspections. Using this method, inspectors directly inspect residential areas to find hazards, track compliance, and promote prompt risk reduction. (BFP, 2022). The program promotes accountability and guarantees that fire safety procedures are tracked and continuously enhanced at the local level by integrating inspections into house calls. Home Fire Safety Visits (HFSVs) or Home Fire Safety Checks (HFSCs) are proactive community-based strategies that fire services use to assess domestic fire risks, increase awareness, and promote preventive behavior(Waring et al., 2024).

During these visits, trained firefighters or community volunteers usually inspect homes for hazards, ensure smoke alarms are in place, and instruct residents on safe electrical practices and

evacuation planning (Hampton et al., 2025). Reducing fire incidents and the injuries and deaths they cause is the goal of several global initiatives. This involves educating and training parents, children, and high-risk groups such as the elderly. (Jonsson et al., 2016) & (Wuschke et al., 2013). These personal visits provide customized guidance and equipment fittings based on the property layout and obvious hazards. The realization that socially disadvantaged people frequently lack the funds to purchase fire safety equipment is one of the factors driving this. In a similar vein, elderly or disabled people might have trouble setting up equipment or require more specialized guidance on potential hazards and escape routes (Jonsson et al., 2016).

(O. Ferrer, 2021) Noted that fire safety inspections and assessments were carried out correctly in Aparri. However, some establishments still broke the Fire Code, creating hazards. As a result, authorities advised BFP Aparri to monitor non-compliant buildings more closely.

Zonal Interventions

The zonal interventions concept under the Bureau of Fire Protection's (BFP) Oplan Ligtas na Pamayanan (OLP) framework represents a systematic, community-based approach to fire prevention and preparedness. According to the 2023 Revised OLP Guidebook, zonal intervention is the process of assigning barangays or communities to one of four fire vulnerability zones—red, orange, yellow, or blue—based on the risk levels determined by the BFP, (2022) and the Community Fire Protection Plan (CFPP).

This zonal classification decides how often and how intensely interventions are carried out, such as hazard removal, evacuation drills, fire safety education, and house-to-house fire safety surveys. OLP enables the BFP to effectively allocate resources and ensure that communities at high risk receive more stringent fire prevention measures by classifying areas according to their vulnerability.

The Guidebook states that the Red Zone indicates the most vulnerable areas, such as those with poor housing conditions or a high frequency of fire incidents. Twice a year, fire officers and community volunteers must perform house-to-house fire safety surveys in these areas, along with hands-on hazard removal and informative seminars. While the Yellow Zone, which has low fire vulnerability, needs interventions every two years, the Orange Zone, which has been classified as moderately vulnerable, requires surveys and lectures once a year. The Blue Zone, which is considered reasonably safe, emphasizes maintenance, evacuation planning, and ongoing monitoring. (BFP, 2022). The zonal intervention approach is a strategic plan that spreads fire safety initiatives relatively evenly while focusing on risk. This is shown through an organized, color-coded classification system for all barangays.

Theoretically, risk-based targeting and community-based disaster preparedness frameworks, which support directing resources to areas with concentrated risks, also lend support to the implementation of zonal interventions (Waring et al., 2024) found that home fire safety visits in targeted areas significantly reduced household fire risks and gave good cost-benefit results.

Fire Safety Lectures and Seminars

Systematic programs like those described in the Bureau of Fire Protection's (BFP) Oplan Ligtas na Pamayanan (OLP) Guidebook help to strengthen fire safety initiatives at the home and community levels. Fire safety education is an important part of prevention efforts. Households can learn about fire risks and safety measures through programs like community fire safety surveys, seminars, and lectures, as described in the OLP Guidebook (BFP, 2022). As previously mentioned, education is an essential tool for raising awareness and guiding families toward safer practices, reinforcing the idea that communities with greater knowledge are more fire-hazard-resistant. The OLP emphasizes how campaigns help foster a culture of long-term fire safety. The authorities urge

barangays to carry out zonal interventions, disseminate educational materials, and use social media platforms to increase advocacy through public fire education campaigns (BFP, 2022). These campaigns help the BFP, and the community maintain awareness, promote positive attitudes, and consistently apply preventive measures together.

Households that received structured fire safety education demonstrated significant improvements in knowledge and preventive behaviors (Hassan Tawfik et al., 2019). This suggests that awareness alone can bring about change when combined with favorable attitudes toward safety procedures. Furthermore, (Nilson & Bonander, 2020) emphasize that knowledge heightens perceptions of risk. Families aware of fire risks were more likely to recognize their own vulnerability, which encouraged them to take precautions, such as installing alarms and rehearsing evacuation procedures. (Nanta & Pisueña, 2025) found that barangays that regularly attended fire safety seminars followed local fire codes more closely and showed a stronger sense of community responsibility for preventing fires. This is consistent with the BFP's belief that education empowers people and leads to behavioral change. According to Espartero, (2025), OLP lectures and seminars were considered by the people of Antique Province to be one of the most successful elements of the program because they promoted collaboration with nearby fire stations and improved knowledge of appropriate fire prevention behavior.

Research, however, indicates that safe practices are not always the result of knowledge. (Gilbert, 2021) found that because residents try to control fires on their own rather than following evacuation protocols, many home fires go unreported. Due to perception and mobility issues, older adults frequently adopt preventive measures, such as smoke alarms, but often lack comprehensive knowledge of risky behaviors (Karemaker et al., 2021).

Drills and Simulation

Building community skills and readiness through training and drills is another key part of OLP. The Guidebook (BFP, 2022). describes organized programs like Technical Skills for Responders (TSR) and Basic Skills for Responders (BSR), which teach volunteers and Community Fire Auxiliary Group (CFAG) members practical firefighting and emergency response skills. (BFP, 2022) The (BFP, 2022) emphasizes that simulation exercises and fire drills are important for testing preparedness, especially in high-risk residential areas. These exercises not only improve individual skills but also boost group confidence and coordination during emergencies.

(Nickerson, 2021) found that fire safety training gives communities the skills to respond to fires safely and confidently. Additionally, it imparts preventive knowledge, empowering communities to eliminate risks before they materialize. Both fire safety protocols and procedures for evacuating and remaining safe in the event of a fire should be covered in training.

According to the Guidebook, the purpose of fire drills and simulation exercises is to assess the community's preparedness to handle fire situations. Residents' involvement in evacuation and suppression protocols, the promptness of responses, and the efficiency of coordination among community fire auxiliaries are all evaluated. The OLP Guidebook recommends conducting these drills at least once a year, especially in communities identified as Red Zones or high-risk areas by the OLP's risk mapping. Residents can practice reporting fires, carrying out evacuation plans, and collaborating with firefighters and barangay responders by using a simulation BFP, (2022). Nickerson, (2021) found that fire safety training gives communities the skills to respond to fires safely and confidently. It also provides preventive knowledge, helping communities reduce risks before they happen. Training should include both fire safety rules and procedures for evacuating and staying safe during a fire.

Involving local leaders, schools, businesses, and vulnerable groups like the elderly and children is encouraged by the Fire Safety Educator and the Community Fire Auxiliary Group (CFAG). This teamwork approach makes sure the community knows its responsibilities during emergencies.

The OLP Guidebook also helps teachers adjust drills to match the specific risks in each sitio or purok, such as crowded homes, narrow pathways, or limited access to water. This localized approach enhances the drills' Efficacy and realism.

(V. Cvetkovic, 2020) asserts that citizens should be trained in the use of fire extinguishers in order to increase safety and get them ready for fire. Experts advise keeping flammable materials out of attics, basements, and hallways in homes and buildings as a preventive measure. All electrical appliances should be turned off before leaving the house, especially when the occupants will be away for an extended period. Additionally, it is crucial to regularly maintain evacuation routes.

Fire Preparedness and Planning

Reducing losses while preserving safe built environments and promoting sustainable disaster management techniques is the aim of FDP (Kihila & Kihila, 2017). Records of international fire incidents, such as the London Grenfell Tower fire in 2017 and the Shanghai fire in 2010, demonstrate that preparedness and causes differ across countries (Anyanwu et al., 2016).

The availability of firefighting equipment, training, and building occupants' awareness of its use were the main topics of a Tanzanian survey on fire preparedness. According to the study, a home's proximity to a fire department is vital for prompt emergency response. Márquez Sierra et al., (2012). Fire preparedness includes extinguishers, detectors, sprinklers, exit routes, and drills. It also involves building connections with local fire authorities, ensuring buildings are safe, and

training people to use the equipment Hsiao & Hsieh, (2023); Jaafar et al., (2022). Together, these actions make the community better prepared and more resilient.

1.4 Schematic Diagram

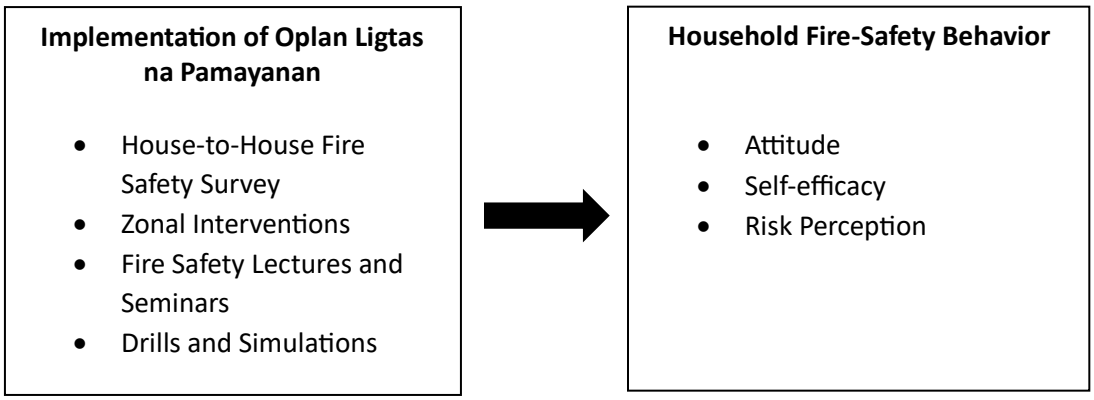


Figure 2. The Schematic Diagram shows the relationship between the dependent and independent variables of the study.

1.5 Statement of the Problem/ Objectives of the Study

This study aims to determine the impact of implementing **Oplan Ligtas na Pamayanan** on the **fire safety behavior of households**. Specifically, it seeks to answer the following questions:

1. **What is the level of implementation in Oplan Ligtas na Pamayanan** in the community in terms of:

    1.1 House-to-House Fire Safety Survey

    1.2 Zonal Interventions

    1.3 Fire Safety Lectures and Seminars

    1.4 Drills and Simulations

2. **What is the level of household fire-safety behavior** of the respondents in terms of:

    2.1 Attitude

    2.2 Self-efficacy

    2.3 Risk Perception

3. **Is there a significant relationship between implementation of Oplan Ligtas na Pamayanan and the household fire-safety behavior of the respondents?**

1.6 Hypothesis

The following hypothesis will guide the study.

H1: There is no significant relationship between the Profile of the Respondents and the implementation of the Oplan Ligtas na Pamayanan and the household fire-safety behavior.

1.7 Definition of Terms

Attitude. Refers how positively the respondents think and feel about fire safety. It is measured by their scores on statements that show their willingness to follow fire prevention rules and join fire safety activities.

Bureau of Fire Protection (BFP). Refers to the local fire officers in BueNasCar. It is measured by their actions in conducting fire inspections, drills, and fire safety education activities under OLP.

Campaigns. It means the fire safety activities done in the community. They are measured by how many people attended, how active they were, and if they practiced fire safety afterward.

Community Fire Auxiliary Group (CFAG). It means barangay volunteers trained by the BFP. It is measured by how often they help in fire prevention, fire response, and OLP activities in the community.

Drills & Simulations. It means practice activities to prepare people for a fire. They are measured by how many people joined and how well they followed the safety steps.

Fire Safety Behavior. Refers to what households in BueNasCar do to prevent fire and stay safe. It is measured by their awareness of fire risks, safe use of electricity, joining fire drills, and being ready for emergencies, like having fire extinguishers and knowing exit routes.

Fire Safety Education. It means teaching people and households about fire safety. It is measured by whether they attend fire safety talks or trainings and how much they learn about preventing fires and staying safe.

Fire Safety Knowledge. The level of awareness and understanding of the respondents regarding fire hazards, preventive measures, and appropriate responses during fire emergencies.

Fire Safety Lectures and Seminars. It means the lessons and talks given by BFP officers to teach people about fire safety. It is measured by the number of sessions attended by participants and their understanding of fire prevention and safe practices.

Households. It means the families in Buenavista, Nasipit, and Carmen who joined the Oplan Ligtas na Pamayanan program. They are measured by their participation in fire safety activities and practices at home.

House-to-House Fire Safety Survey. It means visits by BFP personnel to each household to check if they follow basic fire safety rules. It is measured by how many households are inspected and the level of compliance with safety standards.

Inspections. It means checking households, businesses, or community areas for fire hazards. It is measured by the number of places checked and how well they follow fire safety rules.

Preparedness. Refers to how ready households are to prevent and handle fire incidents. It is measured by whether they have fire safety tools, a plan for emergencies, and if they join fire safety drills.

Risk Perception. Refers to how aware the respondents are of fire dangers and how serious they think these dangers are. It is measured by their answers to questions about the chances and possible impact of fires at home and in the community.

Self-efficacy. Refers to how confident respondents are in doing actions to prevent fires and respond safely during fire emergencies. It is measured by their answers to questions about their ability to follow fire safety practices

Zonal Intervention. It means the fire safety activities organized by the BFP in specific areas of a barangay. It is measured by the number of activities held and the participation of residents in those zones.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter explains the methods used to achieve the study's goal. It also covers the research materials, the methods for analyzing the data collected, the research sample chosen from the population, and the methodology used as a guide for the study. The study also looked at ethical considerations and carefully reviewed the strengths and weaknesses of the methodology.

2.1 Research Design

This study uses a descriptive–quantitative research design, which, according to (McCombes, 2023), systematically describes and measures the characteristics of a population or phenomenon without changing any variables. Using this method, the impact of the Oplan Ligtas na Pamayanan (OLP) program of the Bureau of Fire Protection (BFP) on households' fire safety practices will be evaluated. The primary data collection tool will be a structured questionnaire using a four-point Likert scale. The degree of respondents' knowledge, readiness, and preventive practices can be systematically measured through this design, which also enables statistical analysis to identify trends and connections.

Households directly exposed to OLP activities, such as community drills, fire safety lectures, and house-to-house inspections, will be purposively sampled. To investigate relationships between program participation and household fire-safety behavior, data will be analyzed using descriptive and inferential statistics, such as frequencies, percentages, weighted means, and, when applicable, Pearson Correlation tests.

2.2 Research Materials and Methods

A questionnaire developed by the researcher to determine households' fire safety practices under the Oplan Ligtas na Pamayanan (OLP) program will be the primary instrument for data collection. To capture three important dimensions—knowledge of fire-safety practices,

household preparedness, and preventive practices the questionnaire will be designed with reference to the Bureau of Fire Protection's OLP Guidebook and pertinent academic literature on community-based fire prevention. (BFP, 2022)

To evaluate respondents' knowledge, frequency of preparedness activities, and consistency in implementing preventive measures, the following sections will include statements rated on a 4-point Likert scale, with response options ranging from 4 (Strongly Agree/Always) to 1 (Strongly Disagree/Never). The final questionnaire will include suggestions from academic reviewers and fire-safety experts from the Bureau of Fire Protection, who will check that each item correctly represents the intended concepts.

2.3 Research Environment, Population and Sample

The study will be conducted in the municipalities of Buenavista, Nasipit, and Carmen (BueNasCar) in Agusan del Norte. These nearby municipalities, which include both urban and rural barangays, have been designated as priority areas for community-based fire prevention initiatives because of their varied housing conditions and vulnerability to residential fire hazards.

The respondents of this study consisted of adult residents of the selected barangays who were directly exposed to the implementation of Oplan Ligtas na Pamayanan (OLP). They were selected using purposive sampling to ensure that the participants had sufficient knowledge and experience regarding fire safety programs in their community.

Specifically, the respondents were 18 years old and above, had been residents of the barangay, and served as household heads or adult household members responsible for home safety. Only those who had participated in or were aware of OLP activities, such as fire safety orientations, house-to-house inspections, or barangay-level fire prevention campaigns conducted by the Bureau of Fire Protection, were included in the study.

Participation in the study was voluntary, and only respondents who expressed willingness to take part and were available during the data-gathering period were considered. These criteria ensured that the data gathered were relevant, reliable, and aligned with the objectives of the research.

To make sure each question properly measures household knowledge, preparedness, and safety practices, the questionnaire will be reviewed by experts from the SMCC College of Criminal Justice Education Department. To ensure clarity and applicability, revisions will be made based on expert input.

To further describe the research environment, the selected municipalities and barangays, along with their respective population and household data, are presented in Table 1.

Table 1. Respondents of the Study

Households	Population	Sample	
	F	f	%
Buenavista			
Barangay 7	253	19	5.3
Barangay 9	402	31	8.7
Barangay10	614	47	13.2
Nasipit			
Barangay 3	247	19	5.3
Barangay 7	67	5	1.4
Sta. Ana	636	48	13.5
Carmen			
Centro	1791	137	38.5
Cahayagan	658	50	14.0
Total	**4668**	**356**	**100.00**

2.4 Research Data and Analysis Plan

To determine the number of households in Buenavista, Nasipit, and Carmen, the researchers will first seek permission from the barangay officials. Following the acquisition of the respondent list, the researchers will meticulously organize the data collection procedure, ensuring that participants' rights, security, and benefits are protected.

The researchers will then give participants an informed consent form that thoroughly explains the goals, methods, and confidentiality precautions of the study. The consent form will be available for community members to read over and sign, attesting to their comprehension and voluntary participation in the study.

After the researchers gather the informed consent forms, the questionnaire will be administered during in-person consultations to accommodate each respondent's preferences. After the questionnaires are distributed, the researchers will gather all of the completed questionnaires and use the selected statistical tool to analyze, total, verify, tabulate, and interpret the results.

After that, all the data will be organized and coded for statistical analysis. Following the analysis, the BFP and the relevant barangay councils will receive a summary of the results to support continued fire safety efforts.

The data will be analyzed using appropriate statistical methods:

Frequency and percentage distribution. To describe the demographic profile of respondents and their fire-safety practices.

Weighted mean. To determine the overall levels of knowledge, preparedness, and preventive practices based on the Likert-scale ratings.

Pearson Correlation. To examine the association between participation in OLP activities and household fire-safety behavior.

Table 1. Weighted Mean Scores Interpretation

Likert/Rating Scale	Mean Range/ Parameter		Description	Remarks/ Interpretation
	Lower Limit	Upper Limit		
4	3.25	4.00	Strongly Agree	High
3	2.50	3.24	Agree	Moderate
2	1.75	2.49	Disagree	Low
1	1.00	1.74	Strongly Disagree	Very Low

2.5 Research Ethical Standards

Prior to data collection, the researchers obtained written consent from the BFP and the relevant barangay councils and ethics clearance from the Institutional Research Ethics Committee. The goals, methods, and possible advantages of the study were explained to each participant. They were told that participation was entirely voluntary and that they could leave at any time without incurring any fees, and they signed a written informed consent form. The researchers kept all gathered information completely private, kept it in password-protected files, and, following the required retention period, disposed of it in compliance with institutional policies.

A. Intellectual Property Rights (IPR)

The research does not involve the development of new technologies, processes, or products that would necessitate patent protection. However, it includes original materials such as reports and surveys that require copyright protection to preserve the integrity of the work and prevent unauthorized reproduction or distribution. Additionally, the study does not involve the creation of any unique brand, logo, or symbol that would warrant trademark registration.

B. Informed Consent

Before the study began, participants were fully informed about its purpose, methods, possible risks, and benefits. The researchers explained the study's goals and procedures clearly to

ensure participants fully understood them. All participants provided written consent via an informed consent form that guaranteed their rights as participants and the confidentiality of their answers. Since no animals or other living organisms were used in the study, ethical approval for animal or living-organism trials was not required.

C. Data Privacy and Confidentiality

Participants' personal information was collected and kept safe following the Philippines' Data Privacy Act of 2012 (RA 10173). To maintain confidentiality and privacy, all participant data was anonymized. Questionnaires and other data were kept in a locked drawer, and digital files were encrypted to block unauthorized access. Only the researchers had access to the data. Furthermore, unless participants specifically consent, anonymized data will not be shared with any third parties; any sharing will be for academic purposes only.

D. Voluntary Participation

Every human participant in the study was made fully aware of their right to discontinue participation at any moment without incurring any penalties, and participation was completely voluntary. To ensure ethical compliance and respect for participants' autonomy, the principles of voluntary participation were rigorously adhered to. Since there were no animal subjects or associated experimental procedures in the study, ethical guidelines for the use of animals were not relevant.

E. Minimization of Harm

A risk assessment was conducted to identify and reduce potential physical, psychological, or emotional harm to protect the welfare of human participants. Before beginning the study, participants received comprehensive information about their rights, including the freedom to discontinue participation at any time if they experienced discomfort or distress. Additionally,

participants had access to support services if needed. The researchers did not use any animals in the study, so ethical considerations for animal participants did not apply.

F. Beneficence and Contribution to Knowledge

The study supports the beneficence principle by making a constructive contribution to society and the academic community. To enhance awareness campaigns and encourage responsible safety practices, it examines how the Oplan Ligtas na Pamayanan program affects locals' fire safety behavior. The research's conclusions aim to promote a more proactive, fire-resilient community. Furthermore, participants will have access to the research findings upon request, guaranteeing openness and information exchange.

G. Justice and Fair Participant Selection

To ensure that all participants were included and excluded based on clear, equitable criteria, the study adhered to the principles of justice and fairness in participant selection. Equal opportunity to participate was ensured by including human participants from diverse backgrounds without discrimination. The researchers did not use any animals in this study, so ethical issues regarding animal testing did not apply.

H. Data Integrity and Accuracy

The research ensured data integrity and accuracy by employing standardized data collection and analysis procedures to ensure reliability and validity. All outputs were thoroughly checked to avoid errors and potential biases. Further, the research acknowledged its limitations, including potential biases and limited sample size, to maintain transparency and ensure the integrity of the research process.

I. Transparency and Honesty in Reporting

The study adheres to the principle of transparency in reporting by ensuring that all research findings will be presented truthfully and accurately. The researchers made every effort to

maintain honesty in the interpretation and presentation of results. Furthermore, all copyrighted and patented materials used in the study were cited correctly and acknowledged to uphold academic integrity and give due credit to sources.

J. Use of Patented or Copyrighted Materials

Since no patented technologies were used in the research, the consent of patent holders was not required. To maintain academic integrity and comply with intellectual property laws, the study used copyrighted materials, including software and literature, and obtained the necessary permissions and acknowledgments.

K. Ethical Considerations for Animal and Human Trials

The researchers fully informed all participants of the potential risks and benefits, and the study complied with ethical standards for human research. Ethical clearance for animal use was not required, as no animals were involved in the study. The researchers safeguarded participants' rights and welfare throughout the research process by adhering to ethical standards.

L. Responsible Use of Artificial Intelligence and Related Technologies

To aid the research process, the study used ethical, responsible artificial intelligence tools. The study used Grammarly to ensure grammatical correctness and clarity. ChatGPT was used in the preliminary stage to help organize the researcher's thoughts. All AI-generated content was thoroughly examined and verified to prevent bias or misinterpretation and preserve data integrity. Additionally, throughout the study, the researchers used these technologies to maintain transparency and strengthen ethical standards.

M. Ethical Approval

The study received ethical approval from the Saint Michael College of Caraga Institutional Review Board. The ethics board approved after the researcher fulfilled all requirements and

finished the evaluation process. This ensures that the study complies with accepted ethical guidelines for the use of human subjects in research.

SURVEY QUESTIONNAIRE

Title: Impact of Oplan Ligtas na Pamayanan on Fire Safety Behavior of Households in BueNasCar
Instructions: This questionnaire aims to assess how the implementation of *Oplan Ligtas na Pamayanan* affects household fire-safety behavior. Your responses are confidential and will be used solely for research purposes. Please read each statement carefully and answer honestly.

Part 1. Implementation of Oplan Ligtas na Pamayanan
Instructions: Indicate how often the following activities occurred in your area during the past year by placing a check (✓) on your answer.

Likert/Rating Scale	Description	Remarks/ Interpretation
4	Three or more	The situation described happened three or more time during the past year
3	Twice	The situation described happened twice time during the past year
2	Once	The situation described happened once time during the past year
1	Never	The situation described did not happened during the past year

A. House-to-House Fire Safety Survey	**Three or more 4**	**Twice 3**	**Once 2**	**Never 1**
1. Fire officers or barangay representatives visited our home to check for fire hazard				
2. The fire safety team explained the purpose of their visit before doing the inspection.				
3. We received a copy or verbal explanation of the results of the house inspection.				
4. Fire safety information materials were given to our household during the visit.				

5. Fire officers or barangay representatives returned to follow up after their first visit.				
B. Zonal Interventions				
1. Our area was identified or announced as part of a fire-risk zone.				
2. Fire safety activities or clean-up projects were carried out in our zone.				
3. Barangay officials and fire officers visited our area to conduct zonal interventions.				
4. Fire safety issues in our area, such as narrow pathways or illegal wiring, were addressed.				
5. Follow-up checks, or monitoring, were done after the zonal interventions.				
C. Fire Safety Lectures and Seminars				
1. Fire safety lectures or seminars were held in our community.				
2. Residents in our area were invited to attend these lectures or seminars.				
3. The lectures or seminars were scheduled at times convenient for the community.				
4. Fire officers used demonstrations or visual aids during the lectures.				
5. Handouts or printed fire prevention materials were given during the lectures or seminars.				
D. Drills and Simulation				
1. Fire drills or simulations exercises were conducted in our community.				

2. Residents in our area took part in the fire drills or simulations.				
3. The fire drills included practice with real or mock fire situations.				
4. Organizers explained or reviewed the results of the fire drill afterward.				
5. Additional fire drills or follow-up simulations were conducted after the first one.				

Part 2. Positive Impact of Oplan Ligtas na Pamayanan on Fire-Safety Behavior
Instructions: Please indicate how much you agree with the following statements by placing a check (✓) on your answer.

These statements describe **how the Oplan Ligtas na Pamayanan program has influenced you and your household.**

TABLE ANALYSIS (LIKERT/RATING SCALE INTERPRETTATION)

Likert/Rating Scale	Description	Remarks/ Interpretation
4	Strongly Agree	High
3	Agree	Moderate
2	Disagree	Low
1	Strongly Disagree	Very Low

A. Attitude	Strongly Agree 4	Agree 3	Disagree 2	Strongly Disagree 1
1. Because of Oplan Ligtas na Pamayanan activities, I value fire safety practices more than before.				
2. The program made me realize the importance of preventing household fires.				
3. I have developed a stronger sense of responsibility toward fire prevention at home.				

4. The program encouraged me to promote fire-safety awareness among my family and neighbors.				
5. I now believe that the active participation in community fire-safety programs is essential.				
B. Self-Efficacy				
1. Through the Oplan Ligtas na Pamayanan activities, I became more confident in responding during a fire emergency.				
2. I learned how to properly use fire-safety equipment such as extinguishers or alarms.				
3. The program improved my ability to identify and correct fire hazards in our household.				
4. I feel capable of guiding my family on what to do in case of fire.				
5. I can now take preventive actions immediately after learning from the program.				
C. Risk Perception				
1. The program helped me understand that household fires can happen anytime if we are careless.				
2. I became more aware of the possible consequences of not following fire-safety practices.				
3. I now recognize the importance of reducing risky behaviors that could cause fire.				
4. The program made me more cautious in daily activities that may lead to fire incidents.				

5. I now pay more attention to the safety of electrical wiring, gas use, and open flames at home.				

REFERENCES

Adams, R. M., Eisenman, D. P., & Glik, D. (2019). Community Advantage and Individual Self-Efficacy Promote Disaster Preparedness: A Multilevel Model among Persons with Disabilities. *International Journal of Environmental Research and Public Health 2019, Vol. 16, Page 2779*, *16*(15), 2779. https://doi.org/10.3390/IJERPH16152779

Adams, R. M., Rivard, H., & Eisenman, D. P. (2017). Who Participates in Building Disaster Resilient Communities: A Cluster-Analytic Approach. *Journal of Public Health Management and Practice : JPHMP*, *23*(1), 37–46. https://doi.org/10.1097/PHH.0000000000000387

Addai, E. K., Tulashie, S. K., Annan, J. S., & Yeboah, I. (2016). Trend of Fire Outbreaks in Ghana and Ways to Prevent These Incidents. *Safety and Health at Work*, *7*(4), 284. https://doi.org/10.1016/J.SHAW.2016.02.004

Adeleye, O. I., Odigwe Ajobiewe, T., & Oladipo, T. O. (2020). Fire Disaster Preparedness of Public Buildings in Ibadan Metropolis, Nigeria. In *Open Science Journal*.

Albano, A. S. C. C. J. A. L. M. J. P. E. F. S. V. P. (2024). *Effectiveness of Household Materials As An Alternative For Fire Extinguisher For Class F Grease Fires 2 | PDF | Fires | Firefighting*. https://www.scribd.com/document/815886459/EFFECTIVENESS-OF-HOUSEHOLD-MATERIALS-AS-AN-ALTERNATIVE-FOR-FIRE-EXTINGUISHER-FOR-CLASS-F-GREASE-FIRES-2

Aliesa Mohd Zul, N., Zaiton Wan Sulaiman, W., Mohd Mokhtar, K., Suria Suhaimi, N., Syafiqah Fauzan, N., & Ilyani Rani, S. (2024). Social and Management Research Journal Knowledge, Attitude, and Practice of Fire Safety Systems and Preparedness Among Students at Higher Learning Institutions. *Social and Management Research Journal*, *21*(2), 95–114. https://doi.org/10.24191/smrj.v21i2.28338

Ann, T., Cunanan, R. A., Lagasca, F. L., Janelle, C., Noriega, J., & Cabauatan, R. R. (2022). Issue 2, e001284 Disaster Management. *Malaysian Journal of Social Sciences and Humanities (MJSSH)*, *7*(2), 1284. https://doi.org/10.47405/mjssh.v7i2.1284

Anyanwu, B., Akaranta, O., & Nwaogazie, I. (2016). Evaluation of Fire Safety Management in a Higher Education Institution: A Case Study of University of Port Harcourt. *Archives of Current Research International*, *4*(4), 1–13. https://doi.org/10.9734/ACRI/2016/27684

Ardani, A., & Permana, D. (2019). Evaluation of Community Participation and Fire and Rescue Sub-dept. In Fire Prevention Tanjung Priuk District. *Ilomata International Journal of Social Science*, *1*(1), 20–28. https://doi.org/10.52728/IJSS.V1I1.35

BFP. (2022). *Oplan Ligtas na Pamayanan*. https://region3.bfp.gov.ph/wp-content/uploads/2023/01/OLP-Guidebook.pdf

CAPIÑA, E. D. and L. G. P. (2022). *Librarika: "AN ACTION RESEARCH ON IMPROVING THE IMPLEMENTATION OF "OPLAN LIGTAS NA PAMAYANAN"; FOR EFFECTIVE FIRE PREVENTION IN CONTEMPORARY TIMES". September 2022*. https://ppsc.librarika.com/search/detail/8280948

Chan, E. Y. Y., Lam, H. C. Y., Chung, P. P. W., Huang, Z., Yung, T. K. C., Ling, K. W. K., Chan, G. K. W., & Chiu, C. P. (2018). Risk Perception and Knowledge in Fire Risk Reduction in a Dong Minority Rural Village in China: A Health-EDRM Education Intervention Study. *International Journal of Disaster Risk Science*, *9*(3), 306–318. https://doi.org/10.1007/S13753-018-0181-X/TABLES/4

Cvetkovic, V. (2020). *Disaster Risk Management: Upravljanje rizicima u vanrednim situacijama*. Scientific-Professional Society for Disaster Risk Management. https://pure.unileoben.ac.at/en/publications/disaster-risk-management-upravljanje-rizicima-u-vanrednim-situaci

Cvetkovic, V. (2021). *Fire Safety Behavior Model for Residential Buildings: Implications for Disaster Risk Reduction*.

Cvetkovic, V. M. (2019). *View of Risk Perception of Building Fires in Belgrade*. https://internationaljournalofdisasterriskmanagement.com/Vol1/article/view/6/5

Espartero, J. R. E. & C. R. D. (2025). *View of Oplan Ligtas Na Pamayanan In The Province Of Antique: Its Effectiveness And Impact On Community Fire Safety*. https://theaspd.com/index.php/ijes/article/view/9579/6871

FDMA. (2024). *Fire Prevention Administration in Japan*. https://www.kaigai-shobo.jp/files/fireserviceinjapan_eng/Fire_prevention_administrative_eng.pdf

Fernandez, G., Tun, A. M., Okazaki, K., Zaw, S. H., & Kyaw, K. (2018). Factors influencing fire, earthquake, and cyclone risk perception in Yangon, Myanmar. *International Journal of Disaster Risk Reduction*, *28*, 140–149. https://doi.org/10.1016/J.IJDRR.2018.02.028

Fernández-Vigil, M., & Echeverría Trueba, B. (2019). Elderly at Home: A Case for the Systematic Collection and Analysis of Fire Statistics in Spain. *Fire Technology*, *55*(6), 2215–2244. https://doi.org/10.1007/S10694-019-00852-6/METRICS

Ferrer, O. (2021). *(PDF) EXTENT OF THE IMPLEMENTATION OF THE FIRE PREVENTION PROGRAMS OF THE BUREAU OF FIRE PROTECTION APARRI FIRE STATION*. https://www.researchgate.net/publication/359577212_EXTENT_OF_THE_IMPLEMENTATION_OF_THE_FIRE_PREVENTION_PROGRAMS_OF_THE_BUREAU_OF_FIRE_PROTECTION_APARRI_FIRE_STATION

Ferrer, R. A., & Klein, W. M. P. (2015). Risk perceptions and health behavior. *Current Opinion in Psychology*, *5*, 85–89. https://doi.org/10.1016/J.COPSYC.2015.03.012

Floyd, D. L., Prentice-Dunn, S., & Rogers, R. W. (2000). A Meta-Analysis of Research on Protection Motivation Theory. *Journal of Applied Social Psychology*, *30*(2), 407–429. https://doi.org/10.1111/J.1559-1816.2000.TB02323.X

Fu, R., Zhang, C., Kan, R., Liu, Y., & Seo, H. (2024). Virtual reality–based simulation for assessing building fire safety design. *Simulation*, *100*(6), 581–594. https://doi.org/10.1177/00375497231225980

Gandawali, K. M., & Abdul, A. B. (2017). Implementation of Oplan Ligtas na Pamayanan Program: Its Implication to the Community's Fire Prevention Awareness. In *SMC Research Journal* (Vol. 1).

Gautami, C., Prajapati, M., & Khurana, R. (2020). ANALYSIS OF FACTORS AFFECTING FIRE SAFETY MANAGEMENT OF RESIDENTIAL BUILDING: A CASE STUDY. *International Research Journal of Engineering and Technology*. www.irjet.net

GFDRR. (2013). *Learning from Japan's Experience in Early Warning for Natural Hazards | GFDRR*. https://www.gfdrr.org/en/feature-story/learning-japans-experience-early-warning-natural-hazards

Gilbert, S. W. (2021). *Human Behavior in Home Fires*. https://doi.org/10.6028/NIST.TN.2191

Grothmann, T., & Reusswig, F. (2006). People at risk of flooding: Why some residents take precautionary action while others do not. *Natural Hazards*, *38*(1–2), 101–120. https://doi.org/10.1007/S11069-005-8604-6/METRICS

Guevara, P. (2025). *Fire Safety: Understanding Its Importance | SafetyCulture*. https://safetyculture.com/topics/fire-safety

Hahm, S., Knuth, D., Kehl, D., & Schmidt, S. (2016). The impact of different natures of experience on risk perception regarding fire-related incidents: A comparison of firefighters and emergency survivors using cross-national data. *Safety Science*, *82*, 274–282. https://doi.org/10.1016/J.SSCI.2015.09.032

Hamann, C. J., Mello, E., Wu, H., Yang, J., Waldron, D., & Ramirez, M. (2016). Disaster Preparedness in Rural Families of Children With Special Health Care Needs. *Disaster Medicine and Public Health Preparedness*, *10*(2), 225–232. https://doi.org/10.1017/DMP.2015.159

Hampton, S., Anderson, H., Scantlebury, A., & Adamson, J. (2025). Home Fire Safety Visits by the Fire and Rescue Service: a qualitative study of the perspectives of firefighters, advocates and service leaders. *BMC Public Health 2025 25:1*, *25*(1), 1–6. https://doi.org/10.1186/S12889-025-24429-X

Hassan Tawfik, E., Abd El- Rahman Mohamed, A., & Attia Abd Elghaffar Moustafa, A. (2019). The Effect of a Home Fire Safety Program on Fire Safety Knowledge and Behavior of Community-Dwelling Older Adults and their Family Caregivers. *Egyptian Journal of Health Care*, *10*(3), 560–579. https://doi.org/10.21608/EJHC.2020.267726

Hsiao, C. J., & Hsieh, S. H. (2023). Real-time fire protection system architecture for building safety. *Journal of Building Engineering*, *67*, 105913. https://doi.org/10.1016/J.JOBE.2023.105913

Jaafar, A. S., Ishak, N., Jaffar, N., Othman, M. A., Abdul Talib, Y., & Nasaruddin, M. A. E. (2022). User Awareness on Fire Escape in Hospital Building in Perak. *International Journal of Academic Research in Business and Social Sciences*, *12*(8). https://doi.org/10.6007/IJARBSS/V12-I8/14483

Jonsson, A., Runefors, M., Särdqvist, S., & Nilson, F. (2016). Fire-Related Mortality in Sweden: Temporal Trends 1952 to 2013. *Fire Technology*, *52*(6), 1697–1707. https://doi.org/10.1007/S10694-015-0551-5/METRICS

Karemaker, M., ten Hoor, G. A., Hagen, R. R., van Schie, C. H. M., Boersma, K., & Ruiter, R. A. C. (2021). Elderly about home fire safety: A qualitative study into home fire safety knowledge and behaviour. *Fire Safety Journal*, *124*, 103391. https://doi.org/10.1016/J.FIRESAF.2021.103391

Kihila, J. M., & Kihila, J. (2017). Fire disaster preparedness and situational analysis in higher learning institutions of Tanzania. *Jàmbá: Journal of Disaster Risk Studies*, *9*(1), 9. https://doi.org/10.4102/JAMBA.V9I1.311

Kinateder, M. T., Kuligowski, E. D., Reneke, P. A., & Peacock, R. D. (2015). Risk perception in fire evacuation behavior revisited: definitions, related concepts, and empirical evidence. *Fire Science Reviews 2015 4:1*, *4*(1), 1–26. https://doi.org/10.1186/S40038-014-0005-Z

Kurata, Y. B., Ong, A. K. S., Prasetyo, Y. T., Dizon, R. M., Persada, S. F., & Nadlifatin, R. (2023). Determining factors affecting perceived effectiveness among Filipinos for fire prevention preparedness in the National Capital Region, Philippines: Integrating Protection Motivation Theory and extended Theory of Planned Behavior. *International Journal of Disaster Risk Reduction*, *85*, 103497. https://doi.org/10.1016/J.IJDRR.2022.103497

Lagata, L. S., Andujar, C. J., Lloyd Lantaco, E. S., Carlo Manuales, K. A., Allanic, E. A., & Cuevas Jr, J. F. (2022). *Challenges Encountered and Insights of the Bureau of Fire Protection Personnel Towards Responding Fire Incident*. https://doi.org/10.46382/MJBAS.2022.6211

Lagumbay, C., Gaviola, J. H., Quite, J., & Sombice, K. (2023). The Implementation of Fire Prevention Programs by the Bureau of Fire Protection in Barangay. Novaliches Proper in Novaliches, Quezon City: Vol.4, No.1. *Ascendens Asia Singapore – Bestlink College of the Philippines Journal of Multidisciplinary Research*, *4*(1). https://ojs.aaresearchindex.com/index.php/aasgbcpjmra/article/view/13940

Liu, J., Ma, R., Song, Y., & Dong, C. (2023). Developing the Urban Fire Safety Co-Management System in China Based on Public Participation. *Fire 2023, Vol. 6, Page 400*, *6*(10), 400. https://doi.org/10.3390/FIRE6100400

Macarubbo, M. P. , Jr. , & R. G. S. (2022). *Librarika: IMPROVEMENT OF THE IMPLEMENTATION OF OPLAN LIGTAS NA PAMAYANAN (OLP) IN THE MUNICIPALITY OF ALLACAPAN.” September 2022*. https://ppsc.librarika.com/search/detail/8262007?utm_

Manalo, J. A., Matsumoto, J., Takahashi, H. G., Villafuerte, M. Q., Olaguera, L. M. P., Ren, G., & Cinco, T. A. (2022). The effect of urbanization on temperature indices in the Philippines. *International Journal of Climatology*, *42*(2), 850–867. https://doi.org/10.1002/JOC.7276

Mangahas, T. L. S., Casimiro, R. R., Gabriel, A. G., Mangahas, T. L. S., Casimiro, R. R., & Gabriel, A. G. (2018). Economically Challenged Women in Disaster Risk Management: Toward a Resilient Filipino Community. *Open Journal of Ecology*, *8*(1), 42–56. https://doi.org/10.4236/OJE.2018.81004

Marceron, J. E., & Rohrbeck, C. A. (2019). Disability and disasters: the role of self-efficacy in emergency preparedness. *Psychology, Health & Medicine*, *24*(1), 83–93. https://doi.org/10.1080/13548506.2018.1492730

Márquez Sierra, F. J., Rubio-Romero, J. C., & Rubio Gámez, M. C. (2012). Status of facilities for fire safety in hotels. *Safety Science*, *50*(7), 1490–1494. https://doi.org/10.1016/J.SSCI.2012.01.006

Martin’s Electrical and Lighting Company. (2025). *DIY Electrical Dangers: The Hidden Costs That Could Bankrupt You*. https://martins-electrical.com/diy-electrical-dangers-the-hidden-costs-that-could-bankrupt-you/

Mccaffrey, S. (2015). *Community Wildfire Preparedness: a Global State-of-the-Knowledge Summary of Social Science Research*. https://doi.org/10.1007/s40725-015-0015-7

McCombes, S. (2023). *Descriptive Research | Definition, Types, Methods & Examples*. https://www.scribbr.com/methodology/descriptive-research/?utm_source=chatgpt.com

Menya, A. A., & ; K'akumu, O. A. (2016). Inter-agency collaboration for fire disaster management in Nairobi City. *Journal of Urban Management*, *5*(1), 32–38. https://doi.org/10.1016/j.jum.2016.08.001

Muico, E. J. G., Canalija, J. K., Ronald, D., & Lucero, J. (2024). Evaluation of Fire Safety Measures in Barangay San Roque, Municipality of Maco: Basis for Intervention Scheme. *Journal Healthcare Treatment Development*, *4*(02), 1–5. https://doi.org/10.55529/JHTD.42.1.5

Nanta, K. J., & Pisueña, J. J. (2025). Barangay Officials' Role in Fire Safety: Understanding and Implementation. *Journal of Interdisciplinary Perspectives*, *3*(8). https://doi.org/10.69569/JIP.2025.417

Nickerson, R. S. (2021). Looking Ahead: Human Factors Challenges in A Changing World. *Looking Ahead: Human Factors Challenges in A Changing World*, 1–463. https://doi.org/10.1201/9781003064039

Nilson, F., & Bonander, C. (2020). Household Fire Protection Practices in Relation to Socio-demographic Characteristics: Evidence from a Swedish National Survey. *Fire Technology*, *56*(3), 1077–1098. https://doi.org/10.1007/S10694-019-00921-W/FIGURES/2

Nilson, F., & Mankell, A. (2022). Is the perceived risk of fire-related injury and trust in local authorities affected by rescue services' actual ability? *Preventive Medicine Reports*, *30*, 102030. https://doi.org/10.1016/J.PMEDR.2022.102030

Nocete, R. O., & Enteria, O. C. (2020). *Practices and Level of Awareness on the Basic Safety Precautions on Electrical Fire Prevention among Residence in Bislig City, Philippines Practices and Level of Awareness on the Basic Safety Precautions on Electrical*. *15*(6), 7–18. https://doi.org/10.9790/1676-1506010718

Nudzor, H. (2023). *View of From Flames to Fortune by Improving Fire Risk Management in the Case of Ghana*. https://resdojournals.com/index.php/jbeo/article/view/244/229

Opiana, J. A. (2017). *BFP bares activities for Fire Prevention Month | Edge Davao*. https://edgedavao.net/the-big-news/2017/03/bfp-bares-activities-fire-prevention-month/

Paulino, M. A., & Matthew Paulino, L. A. (2023). Causes of Household Fires in Rural Areas: an Exploratory Research. *Asian Journal of Community Services*, *2*(5), 389–394. https://doi.org/10.55927/AJCS.V2I5.4036

Quiroz, N. F., Walls, R., & Cicione, A. (2021). Towards Understanding Fire Causes in Informal Settlements Based on Inhabitant Risk Perception. *Fire 2021, Vol. 4, Page 39*, *4*(3), 39. https://doi.org/10.3390/FIRE4030039

Rahman, M. Z., Atun, F., & Martinez, J. (2021). Earthquake and Fire Hazard Risk Perception: A Study on the Emerging Rangpur City of Bangladesh. *IDRiM Journal*, *11*(1), 1–25. https://doi.org/10.5595/001C.28382

Rivera, J. D. (2020). The likelihood of having a household emergency plan: understanding factors in the US context. *Natural Hazards*, *104*(2), 1331–1343. https://doi.org/10.1007/S11069-020-04217-Z/METRICS

Rogers, R. W. (1975). A protection motivation theory of fear appeals and attitude change. *Journal of Psychology*, *91*(1), 93–114. https://doi.org/10.1080/00223980.1975.9915803

Ronan, K. R., Alisic, E., Towers, B., Johnson, V. A., & Johnston, D. M. (2015). Disaster Preparedness for Children and Families: a Critical Review. *Current Psychiatry Reports*, *17*(7), 1–9. https://doi.org/10.1007/S11920-015-0589-6

Rosales, L. G. (2019). *BFP calls for Fire Safety | BFP-CAR*. https://car.bfp.gov.ph/bfp-calls-for-fire-safety/?fbclid=IwAR0HHG-oq8dwdaI1dSFowBv00-_Q1_5kqNVKbpZSrQ2SxBANq-l0szjtYiU

Samaddar, S., Chatterjee, R., Misra, B., & Tatano, H. (2014). Outcome-expectancy and self-efficacy: Reasons or results of flood preparedness intention? *International Journal of Disaster Risk Reduction*, *8*, 91–99. https://doi.org/10.1016/J.IJDRR.2014.02.002

Senthilkumaran, M., Nazari, G., MacDermid, J. C., Roche, K., & Sopko, K. (2019). Effectiveness of home fire safety interventions. A systematic review and meta-analysis. *PLOS ONE*, *14*(5), e0215724. https://doi.org/10.1371/JOURNAL.PONE.0215724

Sharan, A., & Gaillard, J. C. (2025). Re-Gendering the Sendai Framework for Disaster Risk Reduction: Experiences of Gender Diverse Groups from India and the Philippines. *International Journal of Disaster Risk Science*, *16*(1), 92–102. https://doi.org/10.1007/S13753-024-00612-3/METRICS

Tang, J. S., & Feng, J. Y. (2018). Residents' Disaster Preparedness after the Meinong Taiwan Earthquake: A Test of Protection Motivation Theory. *International Journal of Environmental Research and Public Health 2018, Vol. 15, Page 1434*, *15*(7), 1434. https://doi.org/10.3390/IJERPH15071434

Waring, S., Fielding, J., & Thomas, M. (2024). Examining the effectiveness and economic benefits of home fire safety visits. *Journal of Risk Research*, *27*(11), 1341–1357. https://doi.org/10.1080/13669877.2024.2447261

Wuschke, K., Clare, J., & Garis, L. (2013). Temporal and geographic clustering of residential structure fires: A theoretical platform for targeted fire prevention. *Fire Safety Journal*, *62*(PART A), 3–12. https://doi.org/10.1016/J.FIRESAF.2013.07.003

Yu, J., Sim, T., Qi, W., & Zhu, Z. (2020). Communication with Local Officials, Self-Efficacy, and Individual Disaster Preparedness: A Case Study of Rural Northwestern China. *Sustainability 2020, Vol. 12, Page 5354*, *12*(13), 5354. https://doi.org/10.3390/SU12135354

Zhang, Y., Shen, L., Ren, Y., Wang, J., Liu, Z., & Yan, H. (2019). How fire safety management attended during the urbanization process in China. *Journal of Cleaner Production*, *236*, 117686. https://doi.org/10.1016/j.jclepro.2019.117686